Kaleem Nawaz Khan

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Education

- **Rochester Institute of Technology (RIT)** *Aug. 2022 – Present*
  Ph.D. in Computing and Information Sciences, CGPA: 4.00/4.00 Rochester, NY
  Research Interests: Autonomous Systems, Connected Vehicles, and Networked Systems.
- **FAST-National University of Computer and Emerging Sciences** *Aug. 2015 – Dec. 2017*
  MS in Computer Science, CGPA: 3.68/4.00 Peshawar, Pakistan
- **FAST-National University of Computer and Emerging Sciences** *Sep. 2011 – Jun. 2015*
  BS in Computer Science, CGPA: 3.71/4.00 Peshawar, Pakistan

Work Experience

- **Graduate Research Assistant** *Aug. 2022 – Present*
  Networked Sensing Systems Lab (NSSL), RIT Rochester, NY
  - Developed cooperative perception systems for connected and autonomous vehicles.
  - Co-designed perception, communication, and computing frameworks for reliable autonomous operation.
- **Research Intern** *May. 2024 – Aug. 2024*
  Connected Vehicle Experience Research Lab, General Motors Warren, MI
  - Designed intelligent data offloading mechanisms for connected vehicles leveraging edge and cloud resources.
  - Modeled and evaluated vehicle-to-edge communication using OMNeT++ and Simu5G across network scenarios.
- **Lecturer** *Oct. 2020 – Aug. 2022*
  University of Engineering and Technology, Mardan Mardan, PK
  - Taught courses in Artificial Intelligence, Mobile Application Development, and Algorithms.
  - Served as Career Liaison Officer (CLO), mentoring students and organizing career development workshops.
- **Research Team Lead** *Apr. 2019 – Sep. 2020*
  National Center of Artificial Intelligence (NCAI) Peshawar, PK
  - Led a diverse research team to develop AI solutions for healthcare applications.
  - Applied deep learning techniques for medical screening to assist cardiology and hematology practitioners.
- **Instructor** *Jul. 2015 – Mar. 2019*
  National University of Computer and Emerging Sciences (FAST-NUCES) Peshawar, PK
  - Conducted labs for Programming, Computer Networks, Databases, and Operating Systems.
  - Mentored students in hands-on learning and practical problem solving.

Publications

- **[SenSys'26]** ARC: Accurate, Real-Time, and Scalable Multi-Vehicle Cooperative Perception
  K. N. Khan, F. Ahmad
  Proceedings of the 2026 ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems
- **[MobiSys'24]** VRF: Vehicle Road-side Point Cloud Fusion
  K. N. Khan, A. Khalid, Y. Turkar, K. Dantu, F. Ahmad
  Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services, 2024
- **[CMIG'23]** ImageCAS: A Large-Scale Dataset and Benchmark for Coronary Artery Segmentation Based on Computed Tomography Angiography Images
  A. Zeng, C. Wu, G. Lin, W. Xie, J. Hong, M. Huang, J. Zhuang, S. Bi, D. Pan, N. Ullah, **K. N. Khan**, Tianchen Wang, Yiyu Shi, Xiaomeng Li, Xiaowei Xu
  Computerized Medical Imaging and Graphics, Vol. 109, 2023
- **[Book Chapter'23]** Advanced Deep Learning for Heart Sounds Classification
  M. S. Khan, F. A. Khan, **K. N. Khan**, S. I. Rana, M. A. A. A. Al-Hashemi
  In *Advances in Deep Generative Models for Medical Artificial Intelligence*, pp. 225–248, 2023

- **[SCN'22]** Op2Vec: An Opcode Embedding Technique and Dataset Design for End-to-End Detection of Android Malware
  K. N. Khan, N. Ullah, S. Ali, M. S. Khan, M. Nauman, A. Ghani
  Security and Communication Networks, 2022
- **[Diagnostics'22]** Deep Learning Assisted Automated Assessment of Thalassaemia from Haemoglobin Electrophoresis Images
  M. S. Khan, A. Ullah, **K. N. Khan**, H. Riaz, Y. M. Yousafzai, T. Rahman, M. E. Chowdhury, S. B. Abul Kashem
  Diagnostics, Vol. 12, No. 10, 2022
- **[Physiol. Meas.'21]** Deep Learning Based Classification of Unsegmented Phonocardiogram Spectrograms Leveraging Transfer Learning
  K. N. Khan, F. A. Khan, A. Abid, T. Olmez, Z. Dokur, A. Khandakar, M. E. Chowdhury, M. S. Khan
  Physiological Measurement, Vol. 42, No. 9, 2021
- **[ICAI'21]** Vision-based Human Posture Classification and Fall Detection Using Convolutional Neural Networks
  R. Hasib, **K. N. Khan**, M. Yu, M. S. Khan
  IEEE 2021 International Conference on Artificial Intelligence (ICAI)
- **[ICAI'21]** Evaluation of Preprocessing Techniques for U-Net Based Automated Liver Segmentation
  M. Islam, **K. N. Khan**, M. S. Khan
  IEEE 2021 International Conference on Artificial Intelligence (ICAI)
- **[ICAI'21]** Performance Evaluation of Background Subtraction Techniques for Video Frames
  S. Qasim, **K. N. Khan**, M. Yu, M. S. Khan
  IEEE 2021 International Conference on Artificial Intelligence (ICAI)
- **[ICOSST'21]** Automatic Classification of Heart Sounds Using Long Short-Term Memory
  B. Ahmad, F. A. Khan, **K. N. Khan**, M. S. Khan
  IEEE 2021 15th International Conference on Open Source Systems and Technologies (ICOSST)
- **[ICAI'21]** Computer-Aided Phonocardiogram Classification Using Multidomain Time and Frequency Features
  J. Dastagir, F. A. Khan, M. S. Khan, **K. N. Khan**
  IEEE 2021 International Conference on Artificial Intelligence (ICAI)
- **[ICAI'21]** Automated Assessment of Lymphocytes Using Machine Learning Techniques
  A. Ahmad, A. Ullah, **K. N. Khan**, M. S. Khan
  IEEE 2021 International Conference on Artificial Intelligence (ICAI)

Research Projects

- **FRC: Radar–Camera Fusion for Intersection Awareness** *Nov. 2025 – Present*
  Research Assistant – Networked Sensing Systems Lab RIT, NY
  - A multimodal perception framework that fuses automotive radar and camera data for intersection awareness.
  - Spatial alignment and synchronization techniques to associate radar detections with camera observations.
- **ARC: Accurate, Real-Time, and Scalable Multi-Vehicle Cooperative Perception** *Jun. 2024 – Mar. 2025*
  Research Assistant – Networked Sensing Systems Lab RIT, NY
  - Designed a scalable cooperative perception framework with high accuracy across multiple vehicles.
  - Developed an alignment refinement layer to improve point cloud registration accuracy in cooperative perception.
  - GPU-accelerated grid-based spatial reasoning to reduce alignment latency and enable selective inter-vehicle sharing.
  - Resulted in a publication at ACM/IEEE SenSys 2026.
- **Smart Vehicle-to-Edge Computation and Data Offloading** *May. 2024 – Aug. 2024*
  Research Intern – Connected Vehicle Experience Research Lab General Motors, Warren, MI
  - Developed a framework for intelligent vehicular computation and data offloading to edge and cloud infrastructure.
  - Simulated 5G vehicular networks in Simu5G/OMNeT++ with mobile vehicles, base stations, and edge servers.
  - Evaluated effects of latency, signal quality, and congestion on offloading performance under varying traffic conditions.
- **VRF: Vehicle Road-side Point Cloud Fusion** *Sep. 2022 – Mar. 2024*
  Research Assistant – Networked Sensing Systems Lab RIT, NY
  - Cooperative perception framework for real-time fusion of vehicle and roadside LiDAR data to mitigate occlusions.

- Map-based alignment strategy to independently align vehicle and roadside point clouds to a shared reference frame.
- Demonstrated real-world performance with 20 ms latency and 5 cm positioning accuracy on testbeds.
- Resulted in a publication at ACM MobiSys 2024.

Awards, Grants, and Fellowships

- **Travel Grant**, Rochester Institute of Technology (RIT), to present research at ACM MobiSys 2024, Tokyo, Japan.
- **M.S. Fellowship**, FAST-National University, Peshawar, Pakistan (2015–2017).
- **Prime Minister National ICT R&D Fund Scholarship** for B.S. in Computer Science, Pakistan (2011–2015).
- **Silver Medal**, B.S. in Computer Science, FAST-National University, Peshawar, Pakistan (2011–2015).
- **Stoori da Pakhtunkhwa Merit Certificate**, Government of Khyber Pakhtunkhwa, Pakistan.

Academic Services and Memberships

- **Conference Reviewer:** Shadow Program Committee of EuroSys 2026, MobiSys'26 Artifact Evaluation Committee.
- **Journal Reviewer:** IEEE Network Magazine, IEEE Open Journal of Intelligent Transportation Systems, Transactions on Consumer Electronics, Scientific Reports, International Journal of Machine Learning and Cybernetics, Journal of Information Security and Applications.
- **Membership:** ACM-Student Member, RIT-Doctoral Students Association, RIT-Pak Students Association.
- **Volunteer**, Imagine RIT 2026, contributed to event operations at a campus-wide creativity and innovation festival.

Student Mentees

- Deze Lyu (Ph.D. CIS), 2026
- Audrey Fuller (MS CS), 2026
- Rushil Patel (MS CS), 2026
- Kilian Jakstis (MS CS), 2026
- Ananth Kamath (MS. CS), 2025
- Dharmik D. Patel (MS CS), 2025
- Hrishit Kotadia (MS CS), 2025

Presentations and Talks

- **Conference Talk**, ACM MobiSys'24, International Conference on Mobile Systems, Applications, and Services, Tokyo
- **Research Showcase**, Presented work on cooperative perception at Imagine RIT, RIT's annual innovation and creativity festival.
- **Guest Lecture:** Delivered a guest talk in the CSCI-733 Autonomous Driving Perception course, Spring 2026, Rochester Institute of Technology.
- **Guest Lecture:** Delivered a guest talk in the CSCI-431 Introduction to Computer Vision course, Fall 2023, Rochester Institute of Technology.

Teaching Experience

- **Graduate Teaching Assistant, RIT** — Rochester, NY
  - CSCI-733 Autonomous Driving Perception (Spring 2026), CSCI-739 Topics in Artificial Intelligence (Spring 2024, Spring 2025), CSCI-431 Introduction to Computer Vision (Fall 2023, Fall 2024),
- **Lecturer, University of Engineering and Technology (UET)** — Mardan, Pakistan
  - Artificial Intelligence, Mobile Application Development, and Design and Analysis of Algorithms.
- **Instructor, FAST-National University** — Peshawar, Pakistan
  - Computer Programming, Computer Networks, Databases, and Operating Systems.

Skills Summary

Programming & Tools: Python, C++, C, SQL, ROS, Docker, CUDA, LaTeX
3D Perception: LiDAR, Stereo Vision, Point Cloud Processing, SLAM
3D Reconstruction: NeRF, 3DGS, SfM
Simulation: CARLA, OMNeT++